

Week 5 Lab 1 Report

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Name: 蔡定宇

Library Name: mergesort

1. Build Commands

I used make to build the whole project.

make

Before rebuilding from scratch, I used:

make clean

The Makefile compiles the source files, builds both the static library and shared library, and links the executables.

Commands used internally

For the static library:

```
g++ -Wall -O2 -c main.cpp -o main.o
g++ -Wall -O2 -c Merge.cpp -o Merge.o
g++ -Wall -O2 -c Mergesort.cpp -o Mergesort.o
g++ -Wall -O2 -c PrintArray.cpp -o PrintArray.o
ar -rcs libmergesort.a Merge.o Mergesort.o PrintArray.o
g++ -o main_static main.o libmergesort.a
```

For the shared library:

```
g++ -Wall -O2 -fPIC -c Merge.cpp -o Merge.pic.o
g++ -Wall -O2 -fPIC -c Mergesort.cpp -o Mergesort.pic.o
g++ -Wall -O2 -fPIC -c PrintArray.cpp -o PrintArray.pic.o
g++ -shared -o libmergesort.so Merge.pic.o Mergesort.pic.o PrintArray.pic.o
g++ -o main_shared main.o -L. -lmergesort
```

2. Run Commands and Environment Variables

To run the executable linked with the static library:

./main_static

To run the executable linked with the shared library:

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LD_LIBRARY_PATH=. ./main_shared

Environment variable used

I used:

LD_LIBRARY_PATH=.

This tells the system to search the current directory for the shared library file libmergesort.so at runtime.

3. Other Worth Mentioning

This lab helped me understand the difference between:

- compiling source files into object files
- building a static library (.a)
- building a shared library (.so)
- linking executables with libraries

I also used -fPIC when building the shared library, because position-independent code is required for .so files.

During compilation, I got this warning in PrintArray.cpp:

warning: comparison of integer expressions of different signedness

This warning did not stop the program from compiling or running correctly.

Both executables produced the correct output:

original:

5 3 8 6 2 7 1 4

sorted:

1 2 3 4 5 6 7 8

4. Feedback

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This homework was useful because it made the difference between static linking and shared linking much more concrete. Writing the commands manually first and then putting them into a Makefile helped me understand the build process more clearly.